LAB ASSIGNMENT - 01

Experiment Title:

Study of LAN Topology, Network Devices and Address Configuration.

Questions:-

1. Draw the Network Topology:

- Identify and draw the LAN topology of your organization.
- Clearly label all components such as router, switch, Access points and end devices (PCs, laptops etc)

Solution:- The Academic campus network uses a hybrid hierarchical topology that combines three distinct structural topologies

It combines:

- Tree Topology
- Star topology
- Ring topology.

This network is further divided into three-layered network model which are

- core layer
- Distribution layer
- Access layer

i) Core layer:

- Location: Server room (Room 210, 2nd floor)
- Equipments:

i) Router and firewall:-

Manages external ISP connections, security filtering and external gateway routing.

ii) Layer-3 (3+) Switches:-

These switches are used in ring loop.

ii) Distribution layer :

The main function is to interconnect the access floor layer and enforces local routing

It consists of 4 distribution switches :-

i) R7 (Room 213, 2nd floor) : Direct link to servers room.
ii) R1 (Room 301, 3rd floor) : Aggregates upper floor traffic
iii) R2 (Room 201, 2nd floor): Interconnected with R1 and R7
iv) R6 (Room 113, 1st floor) : Aggregate lower floor and ground floor units.

Components used :-

- TP-Link TL-SG1218MPE
- TP-Link Omada SG3428XMP

iii) Access layer :

The Access layer directly connects end-user devices.

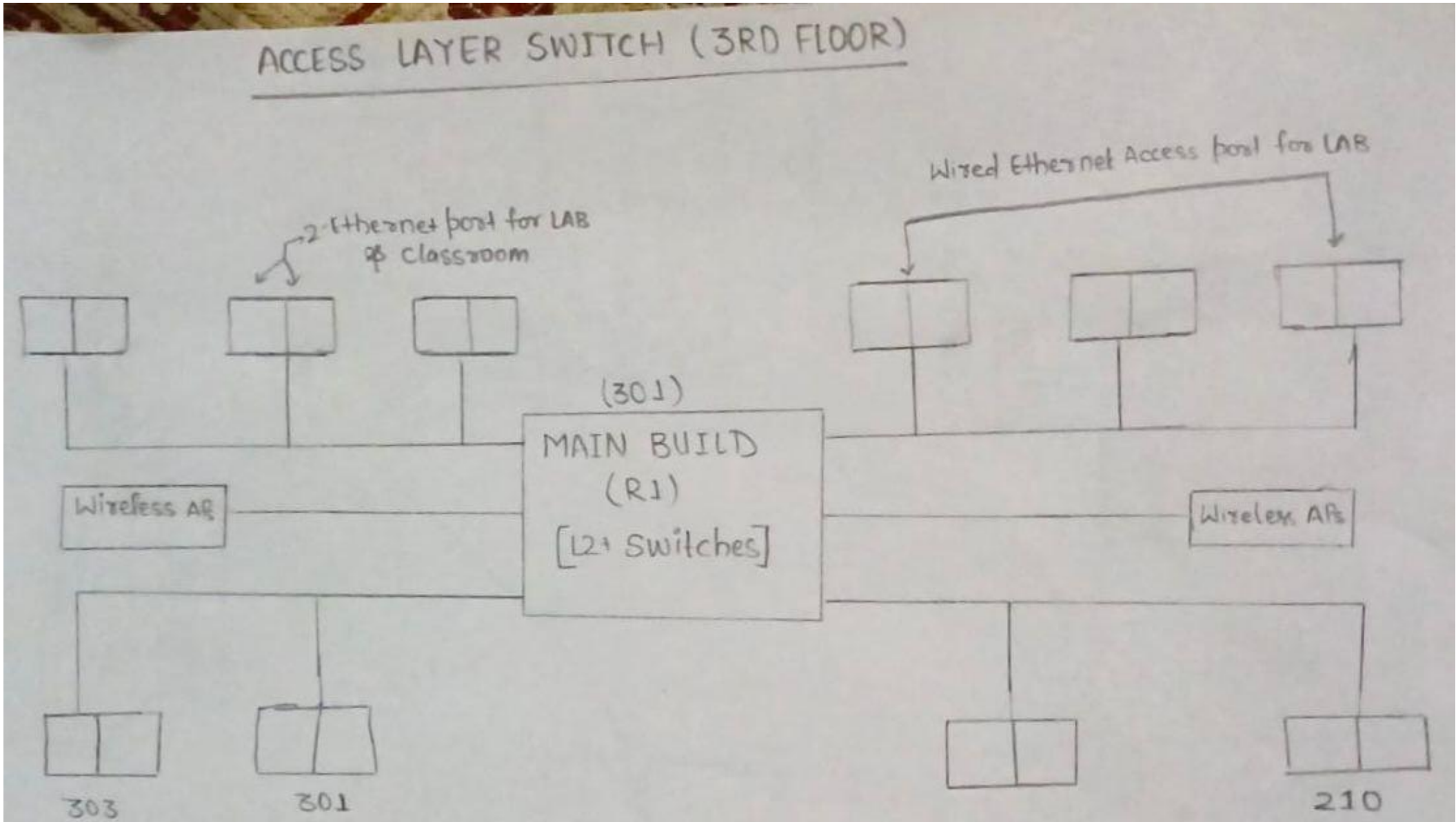
Connected Devices :-

- Classroom Ethernet ports
- Laboratory systems
- Faulty Room PCs
- Wireless Access points (APs)
- outdoor wireless APs

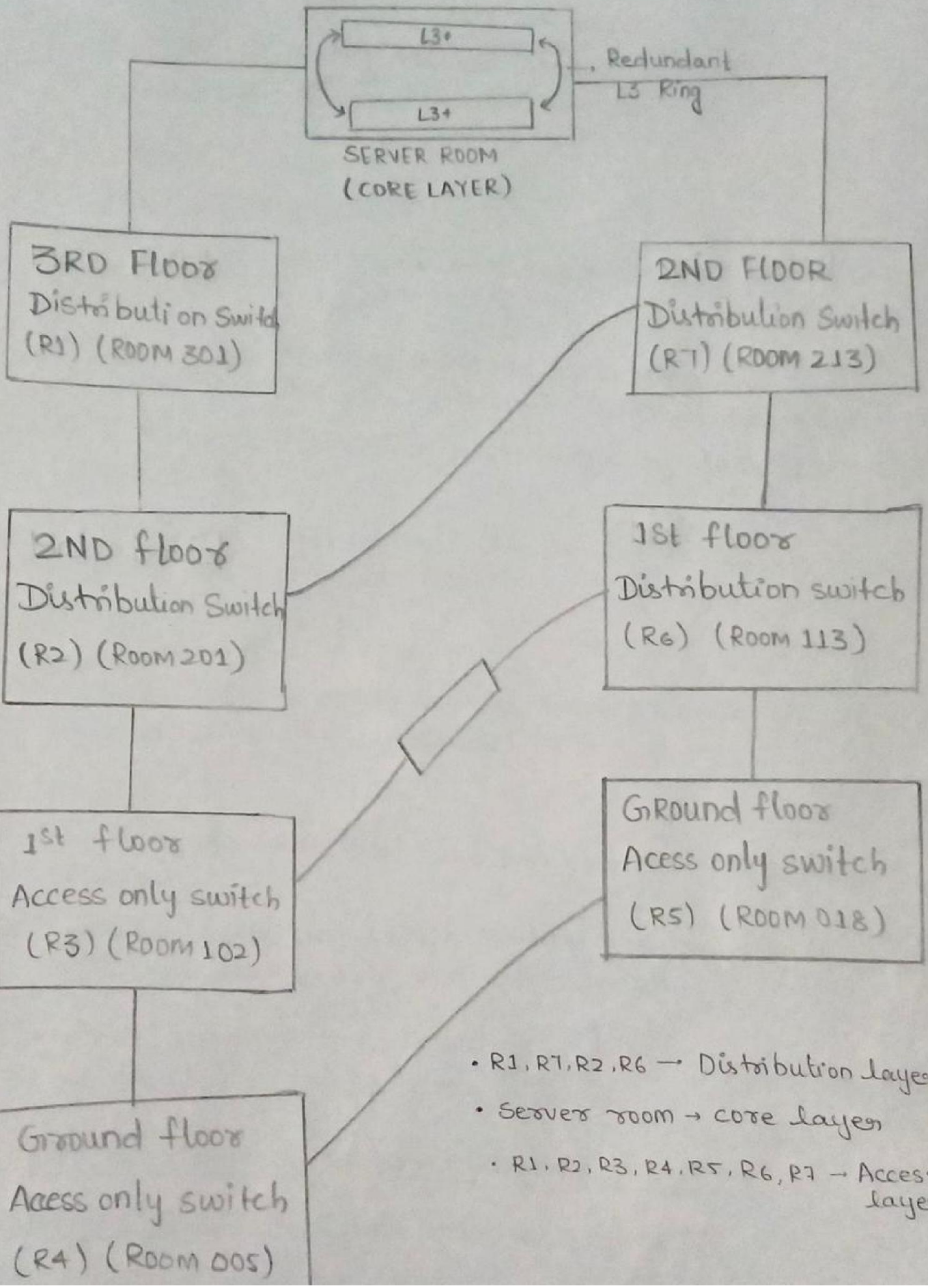
Access layer Switches :-

- R1 → 3rd floor
- R2 → 2nd floor
- R3 → 1st floor
- R4 → Ground floor
- R5 → Ground floor
- R6 → 1st floor
- R7 → 2nd floor

ACCESS LAYER SWITCH (3RD FLOOR)



ACADEMIC NETWORK TOPOLOGY



- R1, R7, R2, R6 → Distribution layer
- Server room → core layer
- R1, R2, R3, R4, R5, R6, R7 → Access layer

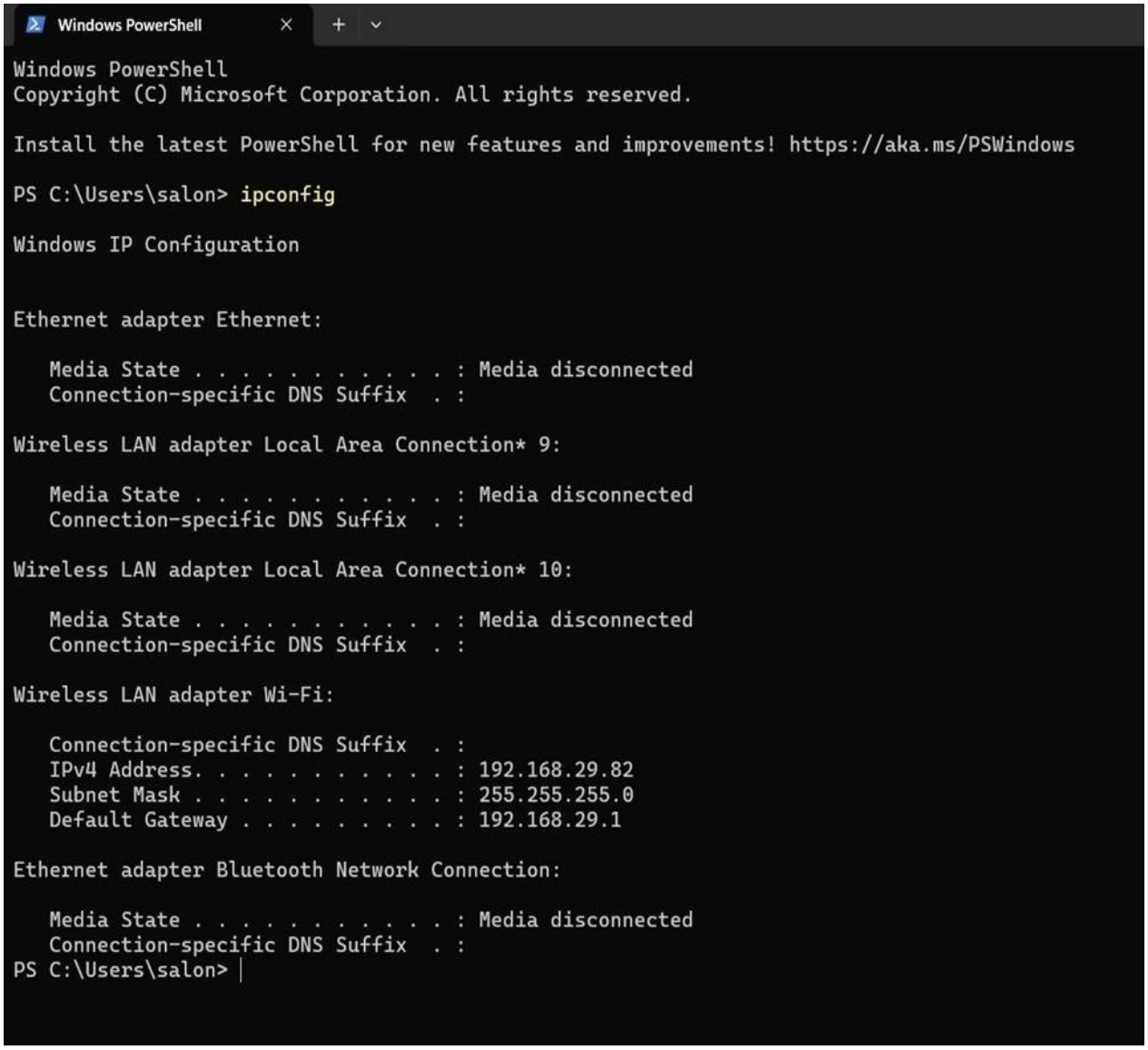
QUESTION 3: Analyze Connection Types:

• Specify whether connections are:

– Wired (Ethernet)

– Wireless (Wi-Fi)

• Mention where each type is used.



->By using ipconfig

QUESTION 4:-

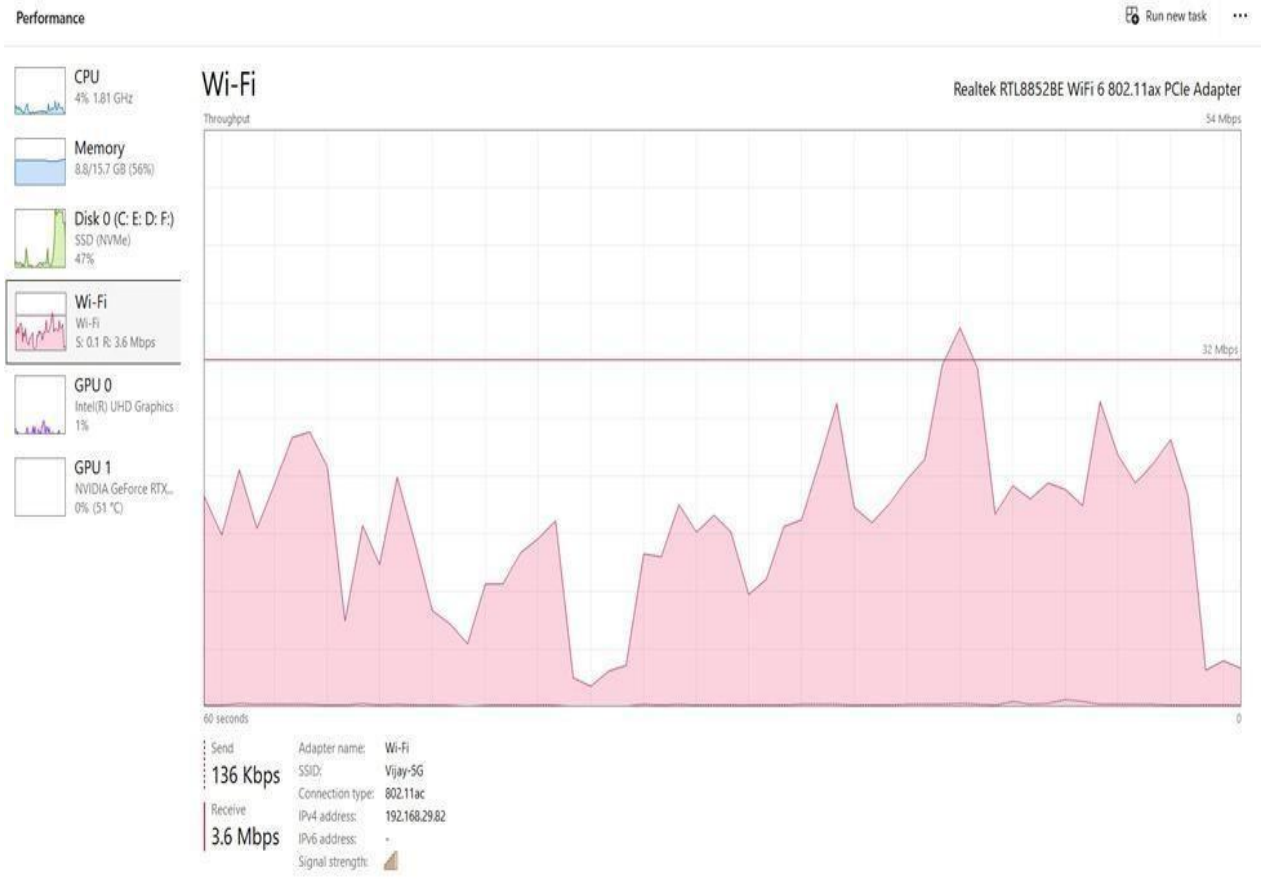
Measure Network Speed:

- Find the link speed of:

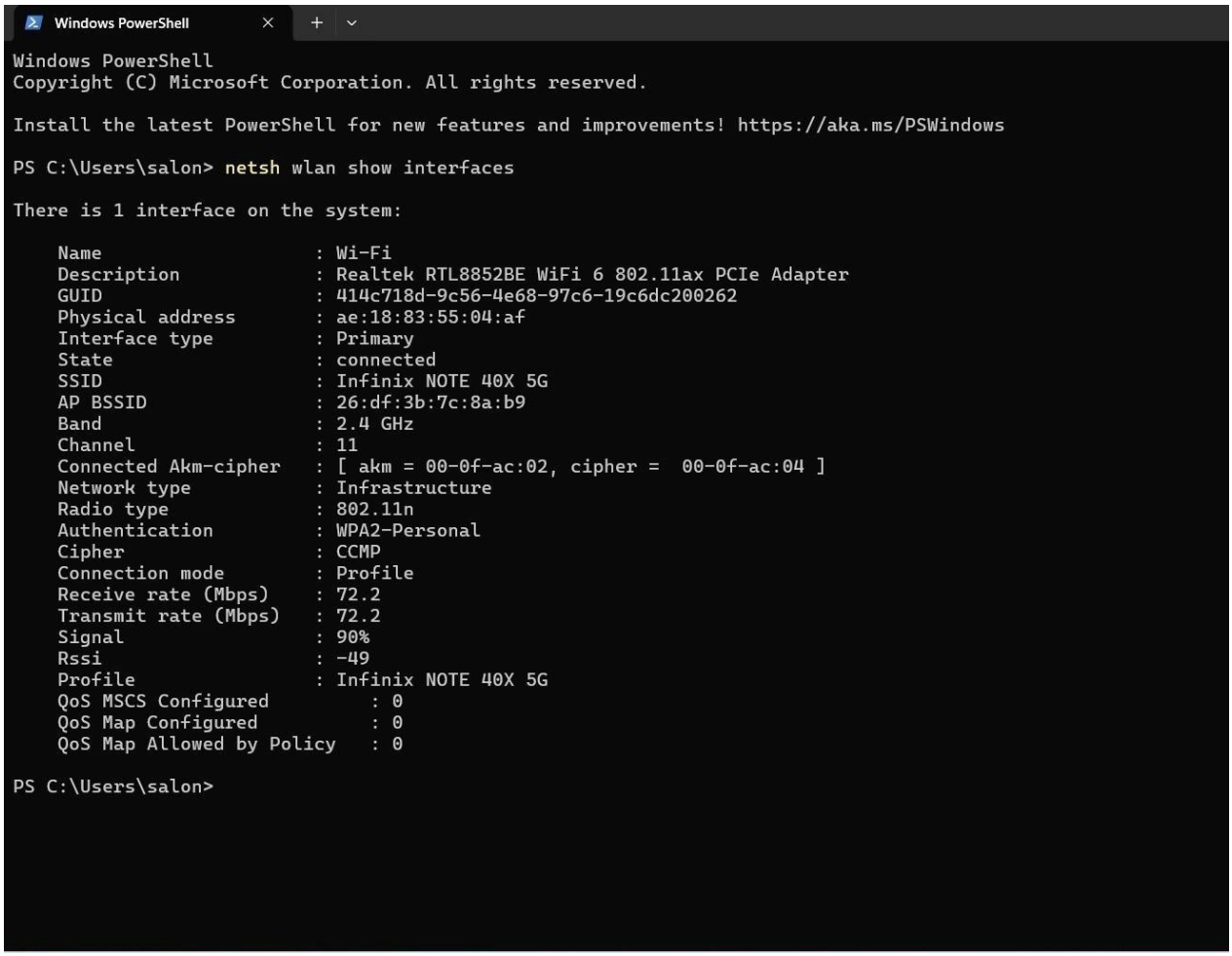
– Wired network (if available)

– Wireless network

- Mention the tools/commands used (e.g., Task Manager, netsh; netsh wlan show interfaces or , etc.).



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->By using netsh wlan show interfaces

QUESTION 5. Find System Network Details:

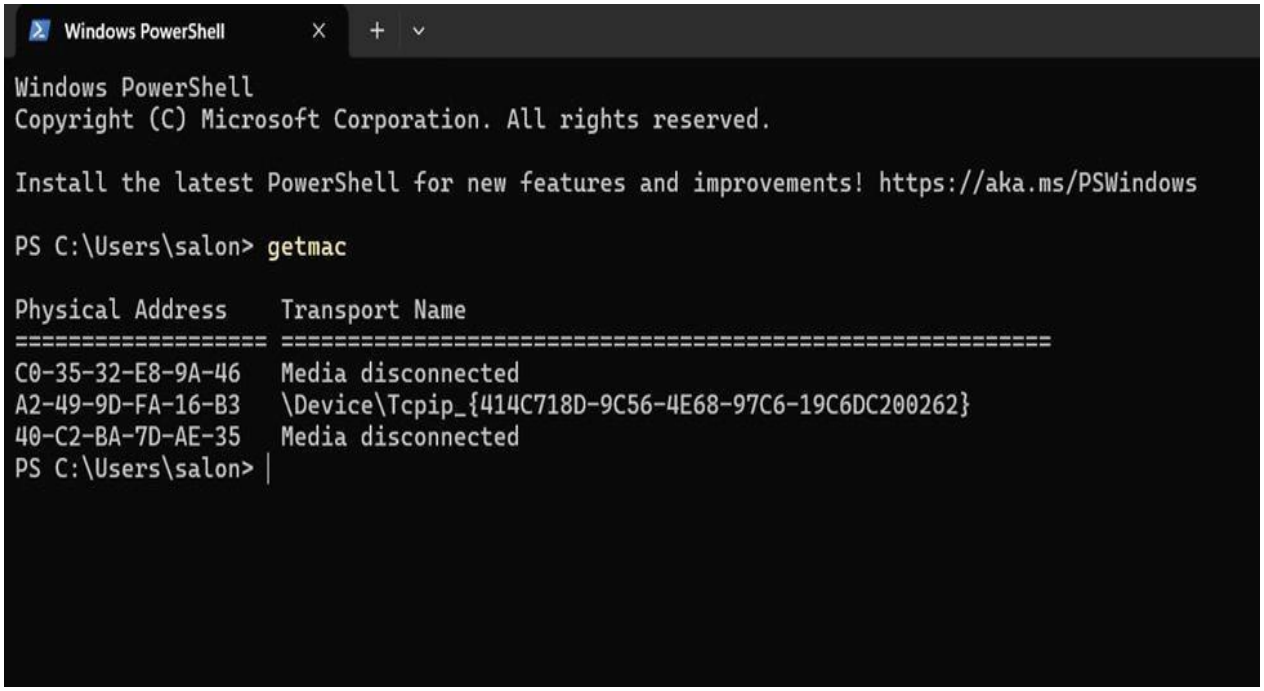
• Determine and record the following of your computer:

– MAC Address

– IP Address (IPv4)

– Subnet Mask

• Mention the command(s) used (e.g., ipconfig, getmac).



->By using getmac

->Ip address using ipconfig.

->subnet mask using ipconfig.

Question-02: Identify Network Devices

- List all networking devices used in the LAN
- Mention the function of each devices

Solution:-

1. Router and firewall:-

Location:- Central Server room (Room 210, 2nd floor)

function

(i) Acts as the primary gateway connecting the internal Campus LAN to the external Internet Service providers

(ii) It Directs incoming and outgoing data traffic using IP routing.

(iii) It filters data packets to protect the school networks from unauthorized external threats.

2. Layer 3+ Core Switches:-

Location:- Distributed in central server room it is deployed as a dual-switch redundant pair.

function:-

(i) It forms the main high-speed backbone of the entire campus network.

(ii) Handles faster inter-VLAN routing (routing data between different floor, labs and departments).

(iii) These switches are configured in ring topology loop to provide physical link reduclancy.

3. TP-Link Omada SG3428XMP (Layered 2+ Switches):

Location: Distribution layer nodes: R1, R2, R6, R7

Function:

(i) Aggregates traffic from the local access layer switches and end user devices on each floor before uploading it to the core layer.

(ii) Provides managed layer 2 capabilities like traffic prioritizing (QoS), security policies and network division.

(iii) It supports inline PoE+ (Power over Ethernet) power directly to downline access points through its 24 gigabit ports.

4. TP-Link TL-SG1218MPE:

Location: Access Layer nodes: R3, R4, R5

Functions:

(i) Connects edge devices directly into the school network infrastructure.

(ii) It offers 16 dedicated PoE+ ports specifically optimized to run network peripherals that require both data and power simultaneously.

5. Wireless Access points (APs):-

Locations: Distributed across all floors and outdoor zones.

Functions:-

(i) It Emits radio signals to provide wireless local area network (wi-fi) coverage across classrooms, labs etc.

(ii) Bridges mobile wireless client devices (laptops, smartphones, tablets) back into the physical wired ethernet distribution layer.

Question - 06 : Explain observations:

Briefly explain:

- Difference between wired & wireless speed

	Wired Speed	Wireless speed
Data flow	Wired supports full-duplex transmission (simultaneous two-way of transmission).	Wireless on the other hand operates on half duplex (only one way at a time)
Signal Interference	Zero interference from physical obstacles or other radio devices.	High interference from walls, microwave ovens and neighbouring routes
Signal transmission	Electrical or light pulses inside a shielded cable	Radio frequencies (2.4 GHz, 5 GHz, 6 GHz) through open air.
Latency	Wired paths have minimal delay	Wireless path suffer from signal overhead and packet loss

- Explain role of subnet mask in networking.

Solution:- A subnet mask is a 32-bit number used to manage network traffic routing. It identifies the network & host portion of an IP address.

- Separation:- It splits IP address into two parts
- Host ID:- It identifies the specific device address
- Network ID: It identifies the specific network address
- Routing:- It tells router if a destination is local or is it is external
- Efficiency:- It prevents unnecessary broadcast traffic across networks.